- Explain the properties and application of Nano-materials.
- Determine the sources and classification of air pollution.
- What is the greenhouse effect?
- What are the sources, ill-effects and remedies of oxides of nitrogen pollutant?
- Write short note on solar cell.
- What are nanomaterials? Write their classification.
- Calculate the pH of the cell: $Pt_{H2(g,1atm)}$ / H^+ || $Ag^+{}_{aq(1.0M)}$ / $Ag_{(s)}$. (Given: E = 1.2 V at 25°C, $E°_{Ag+/Ag}$ = +0.8 V).
- Write short note on electrostatic precipitator used to control particulate matter.
- What is nanomaterial? How are they different from bulk material?
- Explain construction and working of Dry cell.
- What are the sources of water pollution?
- For the cell: $Zn_{(s)}$ / $Zn^{2+}{}_{(aq)}$(1M) || $Fe^{2+}{}_{(aq)}$(1M) / $Fe^{3+}{}_{(aq)}$(0.5M). Given: E°(Zn^{2+}/Zn) = -0.76 V and E°(Fe^{3+}/Fe^{2+}) = 0.77 V. Write the cell reaction and calculate the emf of the cell at 25°C.
- Write a short note on acid rain.
- Explain construction and working of H_2-O_2 fuel cell.
- What is the difference between the top down and bottom up approach of synthesis of nanomaterial? Write classification of nanomaterials.
- What is electromotive force (Emf) of the cell? For the cell: $Cr_{(s)}$ | $Cr^{3+}{}_{(aq)}$(0.1M) || $Fe^{2+}{}_{(aq)}$ (0.01M) | $Fe_{(s)}$. Given: E°(Cr^{3+}/Cr) = -0.74 V and E°(Fe^{2+}/Fe) = -0.44 V. Write the cell reaction and calculate the emf of the cell at 25°C.
- What are the sources, ill effects and remedies of CO & organic vapours air pollutants?

Unit 4: Stereochemistry & Reaction Mechanism

- Assign the absolute configuration R or S in the given compounds. *(Structures provided in exam paper)*
- Write the difference between the inductive effect and the mesomeric effect.